ELEC40001 -A- MATHEMATICS: PRINCIPLES AND ENGINEERING APPLICATIONS -SOLUTIONS 2020

1. [Q1. A,B,C,D: SEEN SIMILAR. E: BOOKWORK]

a) Express in the form $x+iy$:

(i) $\left(\dfrac{3i-2}{2i+3}\right)^2$,

SOLUTION

$\dfrac{-2+3i}{3+2i}\left(\dfrac{3-2i}{3-2i}\right)=\dfrac{13i}{13}=i$, so squaring gives the answer as -1.

(ii) $\left(\dfrac{1-i}{\sqrt{2}}\right)^{2021}$,

SOLUTION

Write as $\dfrac{1}{\sqrt{2}}-i\dfrac{1}{\sqrt{2}}=e^{-i\frac{\pi}{4}}$ so that

$$\left(\frac{1-i}{\sqrt{2}}\right)^{2021}=e^{-i2021\left(\frac{\pi}{4}\right)}=e^{-i\left(505\pi+\frac{\pi}{4}\right)}=-e^{-i\frac{\pi}{4}}=-\frac{1}{\sqrt{2}}+i\frac{1}{\sqrt{2}}$$

(iii) $(-1)^i$,

SOLUTION

Write $-1=e^{i(\pi+2n\pi)}$ so that $(-1)^i=\left(e^{i(\pi+2n\pi)}\right)^i=e^{-(\pi+2n\pi)}$, infinitely many solutions, though the principal one, $e^{-\pi}$, is acceptable.

b) (i) Obtain the real and imaginary parts of $\cos(iz)$; [3]

SOLUTION Begin with

$$\cos(iz)=\cos[i(x+iy)]=\cos(-y+ix)$$

and use the trigonometric identitiy for $\cos(A+B)$ to write

$$\cos(iz)=\cos(-y)\cos(ix)-\sin(-y)\sin(ix)=\cos y\cosh x+i\sin y\sinh x\,,$$

so that
$\mathrm{Re}[\cos(iz)] = \cos y \cosh x$, and
$\mathrm{Im}[\cos(iz)] = \sin y \sinh x$.

(ii) Hence, or otherwise, obtain all complex z satisfying the equation

$$\cos(iz) = -i \, .$$

SOLUTION Rewrite as

$$\cos y \cosh x + i \sin y \sinh x = -i$$

and equate real and imaginary parts:
(i) $\cos y \cosh x = 0$; (ii) $\sin y \sinh x = -1$.

From (i) we get either $\cosh x = 0$ which has no solutions, or $\cos y = 0$. The zeroes of the cosine function are the odd multiples of $\pi 2$ so

$$y = (2n+1)\frac{\pi}{2} \, .$$

Check this in (i), where

$$\sin[2n+1)\pi/2] = \pm 1$$

so we distinguish

$$y = \frac{\pi}{2} + 2n\pi \Rightarrow \sin y = 1 \Rightarrow \sinh x = -1 \Rightarrow x = \sinh^{-1}(-1) = -\sinh^{-1}(1)$$

and

$$y = -\frac{\pi}{2} + 2n\pi \Rightarrow \sin y = -1 \Rightarrow \sinh x = 1 \Rightarrow x = \sinh^{-1}(1)$$

giving solutions

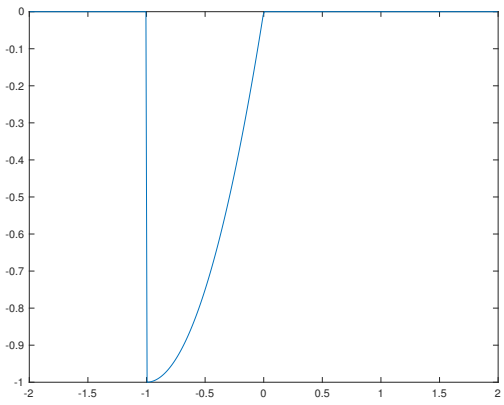
$$z = \pm\left(\sinh^{-1}(1) - \frac{\pi}{2}i\right) + 2n\pi i \, ,$$

where n is any integer.

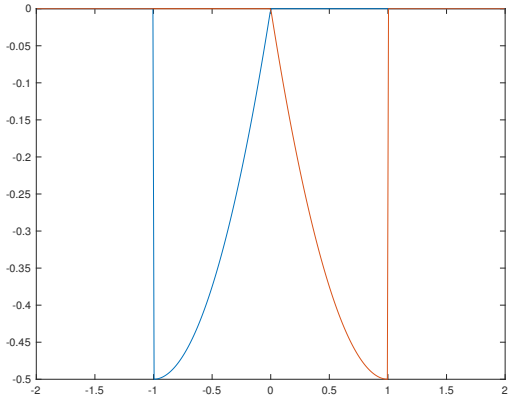
c) Sketch the function $f(t) = t(2+t)[H(t+1) - H(t)]$, where $H(t)$ is the Heaviside function. Hence sketch $f_e(t)$ and $f_o(t)$, the even and odd parts of $f(t)$, taking care to explain your reasoning.

SOLUTION The function is a parabola, crossing the $t-$axis at $t = -2$

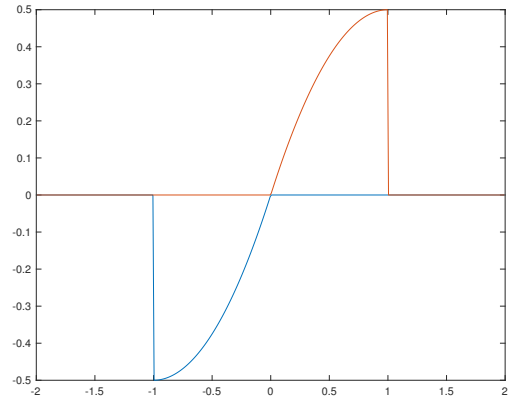
and $t = 0$, with a minimum at $t = -1$. The two Heaviside functions switch on at the minimum and off at the origin, giving the sketch:



To sketch the even part, it's sufficient to realize that $f(-t)$ is simply $f(t)$ reflected in the vertical axis, into the fourth quadrant, and we can add the two graphically, dividing by 2, to give the even part, $f_e(t) = \frac{1}{2}[f(t)+f(-t)]$:



and likewise for the odd part, we need $-f(-t)$ which is $f(t)$ reflected twice, and lies in the first quadrant, giving $f_o(t) = \frac{1}{2}[f(t)-f(-t)]$:



d) Obtain the limits

(i) $\lim_{x\to\pi/2} \dfrac{\cos x}{1+\cos(2x)}$.

SOLUTION

This is a case of "$0/0$", so we can use l'Hopital. Differentiation numerator and denominator, we get

$$\lim_{x\to\pi/2} \frac{\cos x}{1+\cos(2x)} = \lim_{x\to\pi/2} \frac{-\sin x}{-2\sin(2x)}$$

and the limit does not exist, as the numerator is non-zero, but the denominator zero, as $x \to \pi/2$.

(ii) $\lim_{x\to\pi/2} \dfrac{\cot^2 x}{1-\sin x}$.

SOLUTION

Again, "0/0", so use l'Hopital:

$$\lim_{x\to\pi/2}\frac{\cot^2 x}{1-\sin x}=\lim_{x\to\pi/2}\frac{2\cot x(-\frac{1}{\sin^2 x})}{-\cos x}.=\lim_{x\to\pi/2}\frac{1}{\sin^3 x}=1\ .$$

e) Using the definition of derivative as limit, or differentiation from first principles, show that

$$\frac{d}{dx}(\cos x)=-\sin x\ .$$

[You may quote the result for $\lim_{x\to 0}\frac{\sin ax}{x}$.]

SOLUTION

Begin with $y=\cos x$ and

$$\frac{dy}{dx}=\lim_{h\to 0}\frac{\cos(x+h)-\cos x}{h}\lim_{h\to 0}\frac{-2\sin[(2x+h)/2]\sin(h/2)}{h}$$

where we have used the identity $\cos A-\cos B=-2\sin(\frac{A+B}{2})\sin(\frac{A-B}{2})$. Now consider as product of two limits:

$$\frac{dy}{dx}=-2\lim_{h\to 0}\sin(x+h/2)\lim_{h\to 0}\frac{\sin(h/2)}{h}=-2(\sin x)\frac{1}{2}=-\sin x.$$

where we have used $\lim_{x\to 0}\frac{\sin(ax)}{x}=a$ in the last step.

An alternative approach would be to use

$$\cos(x+h)-\cos x=\cos x\cos h-\sin x\sin h-\cos x=\cos x(\cos h-1)-\sin x\sin h$$

and the next step would be to evaluate the limit of $(\cos h-1)/h$.

2. [Q2. A,C,D: SEEN SIMILAR; B,E: BOOKWORK]

a) Differentiate to obtain $\frac{dy}{dx}$: [6]

(i) $y=(\sin x)^x$,

SOLUTION

Take the logarithm:

$$\ln y=\ln(\sin x)^x=x\ln(\sin x)$$

and differentiate implicitly on the left, using product and chain rule on the right:

$$\frac{1}{y}\frac{dy}{dx} = \frac{x}{\sin x}\cos x + \ln(\sin x) \Rightarrow \frac{dy}{dx} = y\left(\frac{x}{\sin x}\cos x + \ln(\sin x)\right) = (\sin x)^x\left(\frac{x}{\sin x}\cos x + \ln(\sin x)\right).$$

(ii) $x^3\sqrt{1-y} - e^{x-y} = 42$.

SOLUTION

Differentiate implicitly:

$$3x^2\sqrt{1-y} + x^3\left(-\frac{1}{2\sqrt{1-y}}\right)\frac{dy}{dx} - e^{x-y}\left(1-\frac{dy}{dx}\right) = 0$$

and collect terms

$$\frac{dy}{dx}\left(e^{x-y} - \frac{x^3}{2\sqrt{1-y}}\right) = e^{x-y} - 3x^2\sqrt{1-y}$$

and finally

$$\frac{dy}{dx} = \frac{e^{x-y} - 3x^2\sqrt{1-y}}{e^{x-y} - \frac{x^3}{2\sqrt{1-y}}} = \frac{e^{x-y}\sqrt{1-y} - 3x^2(1-y)}{e^{x-y}\sqrt{1-y} - \frac{1}{2}x^3}.$$

b) Given $y = \left(\sin^{-1}x\right)^2$, show that $\sqrt{1-x^2}\dfrac{dy}{dx} = 2\sin^{-1}x$.

SOLUTION

Differentiating $y = \left(\sin^{-1}x\right)^2$, we get

$$\frac{dy}{dx} = 2(\sin^{-1}x)\frac{1}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}\frac{dy}{dx} = 2\sin^{-1}x,$$

as required.

Hence, or otherwise, show that $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - 2 = 0$.

SOLUTION

Differentiate $\sqrt{1-x^2}\dfrac{dy}{dx} = 2\sin^{-1}x$ implicitly:

$$\frac{1}{2}\frac{1}{\sqrt{1-x^2}}(-2x)\frac{dy}{dx} + \sqrt{1-x^2}\frac{d^2y}{dx^2} = 2\frac{1}{\sqrt{1-x^2}}$$

and multiply by $\sqrt{1-x^2}$ to get

$$-x\frac{dy}{dx} + (1-x^2)\frac{d^2y}{dx^2} = 2,$$

as required.

c) Evaluate the integrals:

(i) $\displaystyle\int \frac{\sec^2 x}{\tan x}\,dx$,

SOLUTION

Given that $\dfrac{d}{dx}\tan x = \sec^2 x$, we can let $u = \tan x \Rightarrow du = \sec^2 x\,dx$ so that

$$\int \frac{\sec^2 x}{\tan x}\,dx \;=\; \int \frac{1}{u}\,du = \ln u + c = \ln(\tan x) + c\ ,$$

where c is an arbitrary constant.

(ii) $\displaystyle\int_0^1 x\tan^{-1} x\,dx$.

SOLUTION

Integrate by parts:
let $u = \tan^{-1} x \Rightarrow du = \dfrac{1}{1+x^2}dx$ and $dv = xdx \Rightarrow v = \dfrac{1}{2}x^2$ giving

$$\int_0^1 x\tan^{-1} x\,dx \;=\; \left[\frac{1}{2}x^2\tan^{-1} x\right]_0^1 - \frac{1}{2}\int_0^1 \frac{x^2}{x^2+1}\,dx = \frac{1}{2}\tan^{-1}(1) - \frac{1}{2}\int_0^1 1 - \frac{1}{1+x^2}\,dx$$

$$= \frac{1}{2}\left(\frac{\pi}{4}\right) - \frac{1}{2}\left[x - \tan^{-1} x\right]_0^1 \;=\; \frac{\pi}{8} - \frac{1}{2}\left(1 - \tan^{-1}(1)\right) = \frac{\pi}{4} - \frac{1}{2}\ .$$

d) Given that $x^3 + x^2 + x + 1 = 0$ has a zero at $x = -1$, use partial fractions to obtain the indefinite integral

$$\int \frac{2}{1+x+x^2+x^3}\,dx.$$

SOLUTION

Factorize, or use polynomial division, to get

$$1 + x + x^2 + x^3 = (x+1)(x^2+1)$$

and use partial fractions on

$$\frac{2}{1+x+x^2+x^3} = \frac{2}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1} = \frac{A(x^2+1)+(x+1)(Bx+C)}{(x+1)(x^2+1)}$$

equate numerators:

$$2 = A(x^2+1) + (x+1)(Bx+C)$$

and as the above holds for all x, we can choose values of x:
Let $x = -1 \Rightarrow 2 = 2A + 0 \Rightarrow A = 1$,

Let $x = 0 \Rightarrow 2 = 1 + C \Rightarrow C = 1$,
Let $x = 1 \Rightarrow 2 = 2 + 2(B+1) \Rightarrow B = -1$.

Hence

$$\int \frac{2}{1+x+x^2+x^3}\,dx = \int \frac{1}{x+1} - \frac{x}{x^2+1} + \frac{1}{x^2+1}\,dx = \ln(x+1) - \frac{1}{2}\ln(x^2+1) + \tan^{-1}x + c$$

where c is an arbitrary constant of integration.

e) Use a substitution to integrate $\dfrac{1}{\sqrt{x^2+1}}$ and hence, or otherwise, show that

$$\sinh^{-1}x = \ln(x + \sqrt{x^2+1}).$$

SOLUTION

Terms involving $x^2 \pm 1$ are best approached with a hyperbolic substitution. As

$$\cosh^2 u - \sinh^2 u = 1 \Rightarrow \sinh^2 u + 1 = \cosh^2 u$$

we let

$$x = \sinh u \Rightarrow \sqrt{x^2+1} = \sqrt{\sinh^2 u + 1} = \cosh u$$

and

$$dx = \cosh u\, du$$

so that

$$\int \frac{1}{\sqrt{x^2+1}}\,dx = \int \frac{1}{\cosh u}\cosh u\, du = \int 1\, du = u + c = \sinh^{-1}x + c$$

For the second part, the fundamental theorem of calculus has that the derivative of $\sinh^{-1}x$ is $\dfrac{1}{\sqrt{x^2+1}}$, by the integral we've just solved. Hence it's sufficient to show that the derivative of $\ln(x+\sqrt{x^2+1})$ is the same:

$$\frac{d}{dx}\left(\ln(x+\sqrt{x^2+1})\right) = \frac{1}{x+\sqrt{x^2+1}}\left(1 + \frac{x}{\sqrt{x^2+1}}\right) = \frac{1}{x+\sqrt{x^2+1}}\left(\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}}\right) = \frac{1}{\sqrt{x^2+1}}\,.$$

An alternative approach would be to let $y = \sinh^{-1}x \Rightarrow x = \sinh y = \frac{1}{2}(e^y - e^{-y})$ and solve this for $e^y = x \pm \sqrt{x^2+1}$, leading to the same result.

3. [Q3. A,B,D: SEEN SIMILAR; C: PARTLY NEW/SEEN SIMILAR]

a) Use the binomial theorem to find the power series for $y = \dfrac{1}{\sqrt{1-x^2}}$.

Hence, or otherwise, find the first four non-zero terms of the series for $\sin^{-1} x$.

SOLUTION

Use the binomial theorem:

$$\frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2} = 1 - \frac{1}{2}(-x^2) + \frac{-\frac{1}{2}(-\frac{1}{2}-1)}{2}(-x^2)^2 + \frac{-\frac{1}{2}(-\frac{1}{2}-1)(-\frac{1}{2}-2)}{6}(-x^2)^3 + \ldots$$

and we stop there as we have four non-zero terms. Convergence requires $|x| < 1$. Hence

$$\frac{1}{\sqrt{1-x^2}} = 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + \frac{5}{16}x^6 + \ldots$$

Integrating we have

$$\sin^{-1} x = \int \frac{1}{\sqrt{1-x^2}}\,dx = \int 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + \frac{5}{16}x^6 + \ldots\,dx = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \frac{5}{112}x^7 + \ldots + C$$

The constant of integration is found to be zero by evaluating at $x = 0$, so

$$\sin^{-1} x = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \frac{5}{112}x^7 + \ldots$$

which converges for $|x| < 1$.

b) Obtain the Taylor series of

$$y = \frac{\ln(x-1)}{x}$$

about $x = 2$, up to the linear term, and give the remainder term R_1.

SOLUTION

We need y' and y'' only:

$$y' = \frac{1}{x(x-1)} - \frac{\ln(x-1)}{x^2} \Rightarrow y'' = \frac{2\ln(x-1)}{x^3} + \frac{2-3x}{x^2(x-1)^2}$$

Hence $y(2) = 0$ and $y'(2) = \frac{1}{2}$, giving

$$y = y(2) + y'(2)(x-2) + R_1 = \frac{1}{2}(x-2) + R_1$$

where the remainder is

$$R_1 = \frac{y''(c)}{2}(x-2)^2$$

with $1 < x < c < 2$ or $2 < c < x$.

c) The following solution of a first-order ordinary differential equation is incorrect.

The ODE $x\dfrac{dy}{dx} = y\ln\left(\dfrac{y}{x}\right) + y$ *is homogeneous.*

Hence, it can be written as $\left(\dfrac{1}{z\ln z}\right)\dfrac{dz}{dx} = \dfrac{1}{x}$ *where* $z = y/x$.

This is in separable form which gives $\ln(\ln(z)) = \ln(x)$,

and hence the solution $y = Axe^x$, *where A is an arbitrary constant of integration.*

Which was the **first** error made in this calculation? Which was the second error? Now solve the ODE correctly.

SOLUTION

The first error occurs after solving the separable form, which should read

$$\ln(\ln(z)) = \ln(x) + C\,;$$

the error was leaving out the constant of integration.

The second error is in the next line: from

$$\ln(\ln(z)) = \ln(x)$$

the solution would be $y = xe^x$, with no constant A.

The correct solution follows from

$$\ln(\ln(z)) = \ln(x) + C = \ln(kx) \;\Rightarrow\; \ln z = kx \Rightarrow z = e^{kx}$$

where $\ln k = C$.

Finally, $z = y/x$ gives

$$\frac{y}{x} = e^{kx} \;\Rightarrow\; y = xe^{kx}$$

where k is an arbitrary constant.

d) The second-order ordinary differential equation

$$(1-x^2)\frac{d^2y}{dx^2} + 2y = 0$$

can be solved with a series solution. Differentiate the ODE n times, using the Leibniz formula. Show that

$$y^{(n+2)}(0) = (n-2)(n+1)y^{(n)}(0)$$

for $n \geq 0$, where $y^{(n)}(0)$ is the nth derivative of y, evaluated at $x = 0$. Given initial conditions $y(0) = 0$ and $y'(0) = -1$, show that

$$y = -x + \frac{1}{3}x^3 + \frac{1}{15}x^5 + \frac{1}{35}x^7 + \frac{1}{63}x^9 + \ldots$$

SOLUTION

Differentiate the ODE n times, using the Leibniz formula on the first term:

$$(1-x^2)y^{(n+2)}(x) + n(-2x)y^{(n+1)}(x) + \frac{n(n-1)}{2}(-2)y^{(n)}(x) + 2y^{(n)}(x) = 0$$

Let $x = 0$ to obtain

$$y^{(n+2)}(0) - n(n-1)y^{(n)}(0) + 2y^{(n)}(0) = 0$$

and rerrange

$$y^{(n+2)}(0) = [n(n-1) - 2]y^{(n)}(0) = (n-2)(n+1)y^{(n)}(0)$$

as required.

The recurrence formula gives $y^{(n)}(0)$ for $n \geq 0$. With initial conditions we have
$y(0) = y^{(0)}(0) = 0$
$y'(0) = y^{(1)}(0) = -1$
For $n = 0$: $y^{(2)}(0) = -2y^{(0)}(0) = 0$
For $n = 1$: $y^{(3)}(0) = -2y^{(1)}(0) = 2$
For $n = 2$: $y^{(4)}(0) = 0y^{(2)}(0) = 0$
Clearly, all the even terms are zero, we continue only with odd ones:
For $n = 3$: $y^{(5)}(0) = 4y^{(3)}(0) = 2 \times 4$
For $n = 5$: $y^{(7)}(0) = 3 \times 6y^{(5)}(0) = 2 \times 3 \times 4 \times 6$
For $n = 7$: $y^{(9)}(0) = 5 \times 8y^{(7)}(0) = 2 \times 3 \times 4 \times 5 \times 6 \times 8$
and so on. Hence the Maclaurin series is

$$y = y(0) + y'(0)x + \frac{y''(0)}{2!}x^3 + \frac{y^{(3)}}{3!}x^3 + \ldots$$

$$= -x + \frac{2}{3!}x^3 + \frac{2 \times 4}{5!}x^5 + \frac{2 \times 3 \times 4 \times 6}{7!}x^7 + \frac{2 \times 3 \times 4 \times 5 \times 6 \times 8}{9!}x^9 + \ldots$$

$$= -x + \frac{1}{3}x^3 + \frac{1}{15}x^5 + \frac{1}{35}x^7 + \frac{1}{63}x^9 + \ldots \text{ as required.}$$

4. [Q4. A,B,D: SEEN SIMILAR; C: BOOKWORK/SEEN SIMILAR]

a) Solve the second-order ordinary differential equation

$$\frac{d^2x}{dt^2} - 2\frac{dx}{dt} - 3x = 4e^{-t}$$

satisfying the initial conditions $x = 0$, and $\frac{dx}{dt} = 1$, when $t = 0$.

SOLUTION

First solve the homogenous equation, $\ddot{x} - 2\dot{x} - 3x = 0$. Characteristic equation: $m^2 - 2m - 3 = 0 \Rightarrow m = 3, -1$, giving the complementary function

$$x_c = Ae^{3t} + Be^{-t} \ .$$

To find the particular integral, we note that the inhomogenous term $4e^{-t}$ is present in the complementary function, so we need to try

$$x_p = Cte^{-t} \Rightarrow \dot{x}_p = Ce^{-t} - Cte^{-t}t \Rightarrow \ddot{x}_p = -2Ce^{-t} + Cte^{-}t$$

and substituting into the inhomogenous ODE we get

$$e^{-t}[-2C + Ct - 2(C - Ct) - 3Ct] = -4Ce^{-t} = 4e^{-t}$$

so $C = -1$, the particular integral is $x_p = -te^{-t}$ and the general solution is

$$x(t) \ = \ x_c + x_p \ = \ Ae^{3t} + Be^{-t} - te^{-t} \ .$$

Finally, we implement the intial conditions:

$x(0) = Ae^0 + Be^0 - 0 = 0 \Rightarrow A = -B$, and

$$\frac{dx}{dt} = 3Ae^{3t} - Be^{-t} - e^{-t} + te^{-t} \Rightarrow \frac{dx}{dt}(0) = 3A - B - 1 = 1$$

Solve the two equations for $A = \frac{1}{2}$ and $B = -\frac{1}{2}$ and the solution is

$$x(t) \ = \ x_c + x_p \ = \ \frac{1}{2}e^{3t} - \frac{1}{2}e^{-t} - te^{-t} \ .$$

b) A regular cylinder of radius r and height h has volume $V = \pi r^2 h$. Suppose the volume is to be approximated by estimating radius and height. What is the maximum (equal) error we can incur in estimating radius and height, if the error in the volume is to be at most 0.3%?

SOLUTION

We use the differential:

$$dV = \frac{\partial V}{\partial r}\, dr + \frac{\partial V}{\partial h}\, dh = 2\pi rh\, dr + \pi r^2\, dh$$

and divide by V to obtain relative errors:

$$\frac{dV}{V} = (2\pi rh\, dr + \pi r^2\, dh)/(\pi r^2 h) = 2\frac{dr}{r} + \frac{dh}{h}$$

The triangle inequality gives

$$\left|\frac{dV}{V}\right| \leq \left|2\frac{dr}{r}+\frac{dh}{h}\right| = 2\left|\frac{dr}{r}\right|+\left|\frac{dh}{h}\right|$$

If the relative error in V is to be at most 0.003, then we need

$$2\left|\frac{dr}{r}\right|+\left|\frac{dh}{h}\right| < 0.003\ .$$

The relative error in r and h is the same, so we need

$$3\left|\frac{dr}{r}\right| < 0.003\ .$$

and the relative error in these can be at most 0.001 or 0.1%.

c) Polar coordinates (r,θ) in are related to cartesian coordinates (x,y) as follows:

$$x = r\cos\theta\ , \qquad y = r\sin\theta\ ,$$

For a function $f(x,y)$, obtain $\dfrac{\partial f}{\partial r}$ and $\dfrac{\partial f}{\partial \theta}$ in terms of $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$.

Hence, or otherwise, show that

$$\frac{\partial f}{\partial x} = \cos\theta\frac{\partial f}{\partial r}-\frac{\sin\theta}{r}\frac{\partial f}{\partial \theta} \quad \text{and} \quad \frac{\partial f}{\partial y} = \sin\theta\frac{\partial f}{\partial r}+\frac{\cos\theta}{r}\frac{\partial f}{\partial \theta},$$

for $r > 0$.

SOLUTION

The chain rule gives

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial r} = \cos\theta\frac{\partial f}{\partial x}+\sin\theta\frac{\partial f}{\partial y}+0$$

and similarly,

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \theta} = -r\sin\theta\frac{\partial f}{\partial x}+r\cos\theta\frac{\partial f}{\partial y}+0$$

To obtain expressions for $\dfrac{\partial f}{\partial x}$, $\dfrac{\partial f}{\partial y}$ in terms of $\dfrac{\partial f}{\partial r}$, $\dfrac{\partial f}{\partial \theta}$, one could define $r = \sqrt{x^2+y^2}$ and $\theta = \tan^{-1} y/x$ and use the chain rule again. Or one multiplies appropriately:

$$\cos\theta\frac{\partial f}{\partial r} - \frac{\sin\theta}{r}\frac{\partial f}{\partial \theta} = \cos^2\theta\frac{\partial f}{\partial x}+\cos\theta\sin\theta\frac{\partial f}{\partial y}+\sin^2\theta\frac{\partial f}{\partial x}-\sin\theta\cos\theta\frac{\partial f}{\partial y} = \frac{\partial f}{\partial x}$$

Similarly,

$$\sin\theta\frac{\partial f}{\partial r}+\frac{\cos\theta}{r}\frac{\partial f}{\partial\theta} = \sin\theta\cos\theta\frac{\partial f}{\partial x}+\sin^2\theta\frac{\partial f}{\partial y}-\cos\theta\sin\theta\frac{\partial f}{\partial x}+\cos^2\theta\frac{\partial f}{\partial y} = \frac{\partial f}{\partial y}\,.$$

d) Sketch the region of integration and change the order of integration to solve the double integral

$$\int_0^6\int_{x/3}^2 x\sqrt{y^3+1}\,dy\,dx\,.$$

SOLUTION

The region is a triangle in the first quadrant bordered on the left by the $y-$axis, above by the horizontal line $y=2$, and below by the diagonal line $y=x/3$, with corners at $(0,0),(0,2)$, and $(6,2)$.

To change the order of integration to begin integrating with respect to x, the limits are, on the left $x=0$, and on the right $x=3y$. With respect to y, the second integration will have limits $y=0\ldots2$. Hence the integral can be rewritten as

$$\int_{y=0}^2\int_{x=0}^{x=3y}\sqrt{y^3+1}\,dx\,dy = \int_0^2\left[\frac{x^2}{2}\right]_{x=0}^{3y}\sqrt{y^3+1}\,dy = \int_0^2\frac{3}{2}(3y^2)\sqrt{y^3+1}\,dy$$

and letting $u=y^3+1\Rightarrow du=3y^2\,dy$ the limits become $u=1\ldots9$ and the integral is now

$$\frac{3}{2}\int_1^9 u^{1/2}\,du = \frac{3}{2}\left[\frac{u^{3/2}}{3/2}\right]_1^9 = 9^{3/2}-1=26\,.$$